

Tentamen/Exam

Biometrisk Identifiering / Biometric Recognition

Akademi/School: School of Information Technology (ITE)

Datum/Date: 4 January 2024

Kurskod/Course code: DI4025

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Tillåtna hjälpmedel/Allowed aids: Räknare/Calculator

Write your anonymisation code here:

INSTRUCTIONS:

Give your answers in the provided space and return this exam to the caretakers (do not forget to write your anonymisation code in the space provided above!).

If you need more space, use extra sheets (do not forget to write your code as well as "Answer X" on the top of the extra sheets)

Write your answers in a structured and understandable way. **Motivate your answers!**

Good luck!

Maximum score: 38

<u>Grade</u>	<u>Score</u>
Fail:	Below 19
Grade 3:	At least 19
Grade 4:	At least 25
Grade 5:	At least 31

Exercise 1. (5p)

Give your answers in the provided space. If you need more space, use extra sheets.

- a) During the lectures, we spoke about two main types of attacks that can be carried out against biometric systems: Presentation Attacks (PA) and Morphing Attacks (MA). Explain:
- How a Presentation Attack is done, where it happens in the biometric system, and why PA can be problematic for the success of biometric systems **(2p)**
 - How a Morphing Attack is done **(1p)**

ANSWER Presentation Attack:

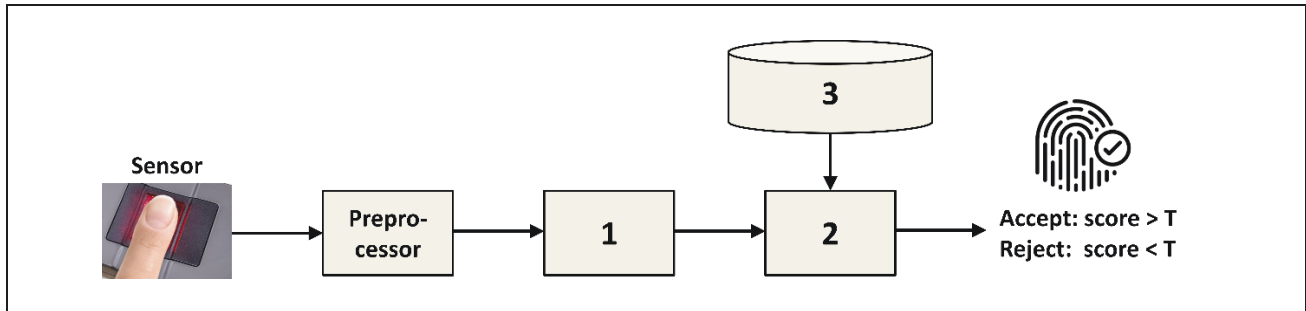
ANSWER Morphing Attack:

- b) During the lectures, we discussed two main sensor technologies for fingerprint acquisition: optical and capacitive. Explain **(2p)**:
- The acquisition principle of each type (i.e. what is measured to capture a fingerprint)
 - Which technology allows a smaller sensor and why

Exercise 2. (4.5p)

Give your answers in the provided space. If you need more space, use extra sheets.

The following image shows the general structure of a biometric system, as discussed in the lectures.



a) Some boxes in the figure have been left in blank, marked as (1), (2), (3). The following is a list of possible candidate names for the boxes left in blank:

- | | | |
|-------------------------|-----------------------|----------------------|
| • Histogram of scores | • Verification system | • Comparator/matcher |
| • Database of templates | • Feature extractor | • False Rejection |
| • EER computation | • Threshold | • Morphing attack |

Choose from the previous list the most appropriate name for each box (1), (2) and (3). **(1p)**
(note: there is only one correct name in the list for each box)

Box 1:

Box 2:

Box 3:

b) Explain the function of each box (1), (2) and (3). **(2.5p)**

An example is given below with the sensor:

- Sensor: captures the biometric data of a user and converts it to a digital signal.

- c) Notice the parameter “T” shown in the right part of the figure. Indicate the name of this parameter and explain its function. **(1p)**

Exercise 3 (12p)

A biometric system is working in VERIFICATION mode. We carry out different genuine and impostor tests. In concrete, 2100 matchings of samples from the same person (genuine) and 1800 matchings of samples from different persons (impostor) are done. In each matching, we employ a SIMILARITY measure, so the HIGHER the value of the measure, the MORE SIMILAR are the samples.

Results of the experiments are reported in the following histograms. The first histogram is when samples from the same person are matched (genuine), and the second histogram is when samples from different persons are matched (impostor). The x-axis of the histogram indicates the similarity value, and the y-axis indicates the number of matchings. A table with results from the histograms is also reported.

Questions

To answer this exercise, there are extra sheets in the next pages that you can separate if you wish (write your code in the sheets if you do so, and do not forget to put them back with your exam).

a) Calculate the FAR and FRR of the system for different thresholds of the similarity measure. Report you results both in:

- -a table, indicating the threshold and the respective errors (employ at least 10 different thresholds, with at least 6 different thresholds where the two histograms overlap)
- -a graph (draw errors in the range 0-30% approximately), indicating clearly in the axes the values of each point

Value: showing the table: **5p**, showing the graph **4p** (total **9p**)

b) Find the approximate threshold of the EER (Equal Error Rate) and the value of the EER in your graph of a).

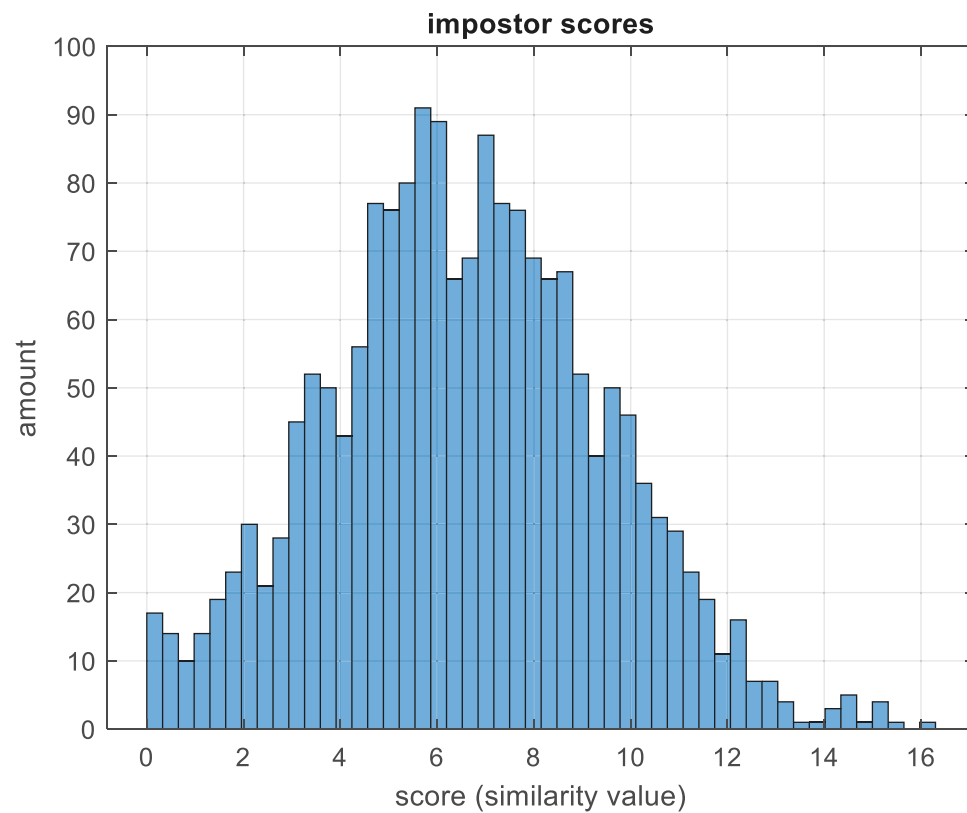
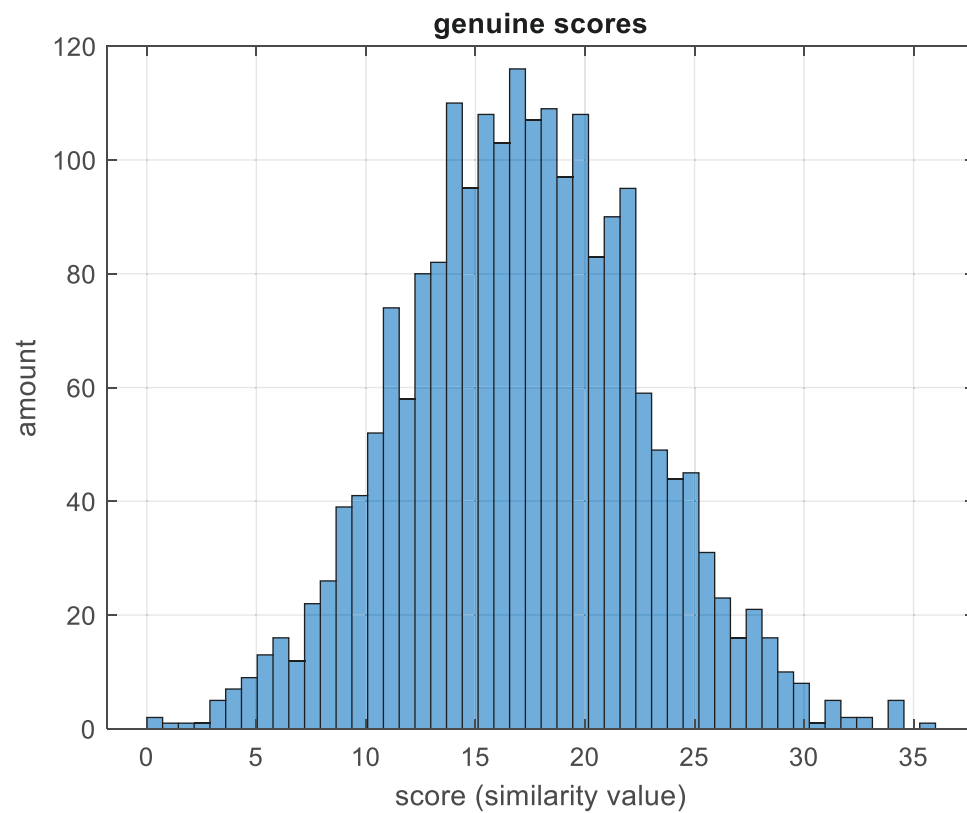
Do not forget to mark the values in your graph of a)!!

Value: **1p**

c) Find the approximate threshold in your graph of a) at which FAR (False Acceptance Rate) is about 4.5% and indicate the value of FRR (in %) at this threshold.

Do not forget to mark the values in your graph of a)!!

Value: **2p**

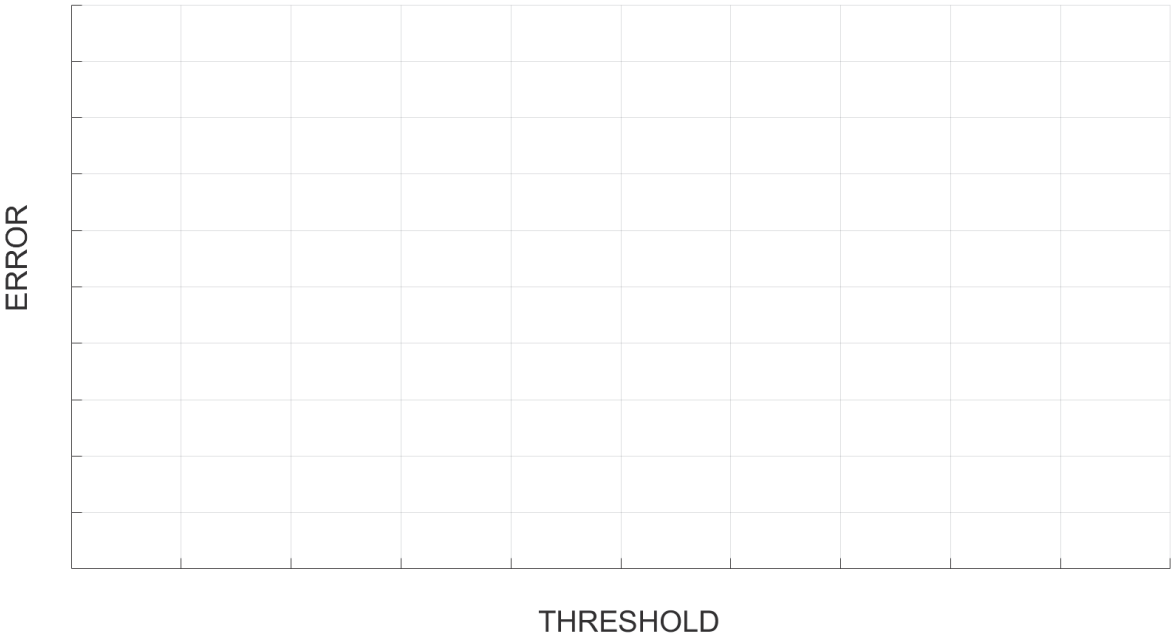


amount genuines	similarity genuines		amount impostors	similarity impostors
2	0.36		17	0.163
1	1.08		14	0.489
1	1.8		10	0.815
1	2.52		14	1.141
5	3.24		19	1.467
7	3.96		23	1.793
9	4.68		30	2.119
13	5.4		21	2.445
16	6.12		28	2.771
12	6.84		45	3.097
22	7.56		52	3.423
26	8.28		50	3.749
39	9		43	4.075
41	9.72		56	4.401
52	10.44		77	4.727
74	11.16		76	5.053
58	11.88		80	5.379
80	12.6		91	5.705
82	13.32		89	6.031
110	14.04		66	6.357
95	14.76		69	6.683
108	15.48		87	7.009
103	16.2		77	7.335
116	16.92		76	7.661
107	17.64		69	7.987
109	18.36		66	8.313
97	19.08		67	8.639
108	19.8		52	8.965
83	20.52		40	9.291
90	21.24		50	9.617
95	21.96		46	9.943
59	22.68		36	10.269
49	23.4		31	10.595
44	24.12		29	10.921
45	24.84		23	11.247
31	25.56		19	11.573
23	26.28		11	11.899
16	27		16	12.225
21	27.72		7	12.551
16	28.44		7	12.877
10	29.16		4	13.203
8	29.88		1	13.529
1	30.6		1	13.855
5	31.32		3	14.181
2	32.04		5	14.507
2	32.76		1	14.833
0	33.48		4	15.159
5	34.2		1	15.485
0	34.92		0	15.811
1	35.64		1	16.137

Answer exercise 3

Table

graph



Space for extra answers of exercise 3 (in case you need it)

Exercise 4 (6p)

A biometric system is working in IDENTIFICATION mode. We carry out a number of tests, with the results shown below. For each test, we obtain a list of ID numbers, sorted by the most similar. The test has been carried out for a total of 16 people, with 3 tests for each person. The individuals have ID numbers: 1, 2, 3, ..., 16

Each line represents a sorted list of ID numbers. Lines are shown in groups of 3, representing the 3 tests of each person: the first group corresponds to ID=1, the next group corresponds to ID=2, and so on.

True ID	Output of the system, sorted by the most similar (first column) to the least similar (last column)															
1	3	4	8	5	1	7	6	2	12	9	15	11	10	13	16	14
1	1	12	2	11	8	14	13	15	10	6	16	9	3	4	5	7
1	1	3	2	6	4	7	13	8	9	11	12	16	15	5	14	10
2	2	5	4	15	13	9	14	12	6	3	10	11	16	8	7	1
2	2	4	3	16	11	7	10	12	9	15	1	13	5	14	6	8
2	9	3	2	5	6	8	4	10	7	14	12	16	13	11	15	1
3	5	4	6	7	9	8	3	11	10	12	13	2	16	15	14	1
3	6	10	4	8	3	7	9	5	15	16	11	2	12	14	13	1
3	7	6	8	9	4	5	3	11	10	16	14	2	13	15	12	1
4	4	8	6	5	16	7	15	14	2	1	10	13	9	12	3	11
4	8	10	6	5	4	9	7	12	3	14	15	11	1	13	16	2
4	7	10	8	9	5	6	4	11	13	3	2	14	12	16	15	1
5	5	6	7	14	11	8	1	4	12	15	13	3	2	10	16	9
5	5	7	3	6	1	8	4	12	16	11	10	2	15	13	14	9
5	6	8	5	9	7	10	13	2	11	3	16	14	15	1	4	12
6	8	6	7	11	9	10	2	16	14	13	3	15	12	4	5	1
6	11	10	6	14	13	9	7	8	12	5	1	15	16	4	2	3
6	8	6	7	4	13	9	11	3	1	14	15	5	16	2	12	10
7	7	9	12	8	11	10	1	15	2	4	13	5	3	6	16	14
7	8	7	9	1	16	3	2	14	4	15	13	12	11	10	5	6
7	7	8	11	9	10	5	13	1	15	16	3	6	14	2	12	4
8	8	13	11	12	2	6	1	16	10	7	9	15	14	3	4	5
8	12	9	8	13	11	10	5	3	16	15	6	2	1	14	7	4
8	11	9	8	12	14	10	13	4	3	16	5	15	1	6	7	2
9	13	9	1	2	15	10	11	16	3	14	12	8	7	5	4	6
9	10	9	12	11	13	2	4	15	14	7	6	3	1	5	8	16
9	9	10	5	12	6	13	11	3	7	4	1	8	2	16	15	14
10	13	11	14	12	15	10	4	9	2	16	3	8	1	7	5	6
10	12	14	16	1	11	10	15	13	2	6	5	3	4	9	7	8
10	12	13	10	16	1	15	14	11	9	4	8	3	2	5	7	6
11	13	11	14	12	7	9	8	5	6	15	16	2	1	3	4	10
11	1	2	16	14	15	12	11	13	3	4	7	6	10	5	8	9

11		12	14	2	15	1	4	11	3	13	16	5	10	9	8	6	7
12		1	14	2	13	16	3	12	15	4	11	7	5	8	9	10	6
12		12	7	10	3	5	4	2	15	9	13	11	8	6	1	16	14
12		13	12	14	11	4	7	3	5	10	16	2	15	8	1	9	6
13		1	4	14	5	2	13	16	15	3	11	8	9	6	10	12	7
13		13	14	15	10	5	8	11	6	12	9	1	7	3	2	4	16
13		15	1	13	14	16	4	2	3	5	6	8	9	11	7	10	12
14		16	4	15	1	14	2	3	7	6	10	13	9	8	5	11	12
14		14	15	16	6	12	5	13	7	1	3	9	2	11	4	10	8
14		15	14	16	3	1	2	4	13	10	11	9	8	12	6	5	7
15		16	15	1	3	9	4	12	14	5	10	8	2	6	11	13	7
15		16	3	2	4	15	1	13	12	11	8	6	14	9	10	7	5
15		1	2	3	6	16	4	5	15	14	11	10	9	7	13	12	8
16		5	1	3	2	16	4	8	14	12	7	15	9	11	10	6	13
16		4	16	6	1	5	2	3	15	10	13	11	9	8	14	12	7
16		4	3	16	1	2	12	14	6	7	8	9	15	11	5	13	10

Questions

To answer this exercise, there are extra sheets in the next pages that you can separate if you wish (write your code in the sheets if you do so, and do not forget to put them back with your exam).

a) Calculate the function $TPIR(M)$ that indicates the probability that the correct identity is given in the first M positions of the sorted list.

Show your calculations of $TPIR(M)$ both in a table and a graph.

Value: showing the table: **3p**, showing the graph **2p** (total **5p**)

b) The system has the requirement that the probability that the correct identity is given within the first 4 positions is at least 71%. Does the system meet this requirement?

Motivate your answer by referring to either the table or graph of a)!!

Value: **1p**

Answer exercise 4

Table

[illegible]

Answer exercise 4

graph



Space for extra answers of exercise 4

Exercise 5 (6p)

The biometric of a person is usually represented as one or more vectors of characteristic (or feature vectors), with each vector of characteristic containing several numbers. Remember the laboratory exercise of hand recognition, where measurements like the finger's length and width, palm width, etc., are put together into a feature vector that represents a description of the hand.

There are several techniques to compare two feature vectors, for example, the Euclidean distance or the normalized scalar product.

The database of our system contains (among others) the feature vectors of two persons db1 and db2:

db1 =

7	-5	0	9	1	2	-1	-2	3	2
---	----	---	---	---	---	----	----	---	---

db2 =

9	4	-1	-1	0	4	4	3	2	-1
---	---	----	----	---	---	---	---	---	----

A person pp1 then should be matched against these two people in the database.

The feature vector of pp1 is:

pp1 =

7	8	0	-3	2	0	-1	-2	5	1
---	---	---	----	---	---	----	----	---	---

Questions

IMPORTANT: You must show **all** your intermediate calculations of a) and b) in the answer sheet, so you demonstrate that you know how these measurements are calculated. Showing only the results will not give any points.

a) Calculate the Euclidean distance between pp1-db1 and pp1-db2, respectively. Show your calculations in the answer sheet and indicate if pp1 is more similar to db1 or db2 based on your results **(2.5p)**

b) Repeat a) but using the normalized scalar product. Show your calculations in the answer sheet and indicate if pp1 is more similar to db1 or db2 based on your results **(3.5p)**

Tip: the normalized scalar product between two vectors a and b is computed as: $\frac{\langle a, b \rangle}{\|a\| \times \|b\|}$, where

$\langle a, b \rangle$ is the scalar product between a and b

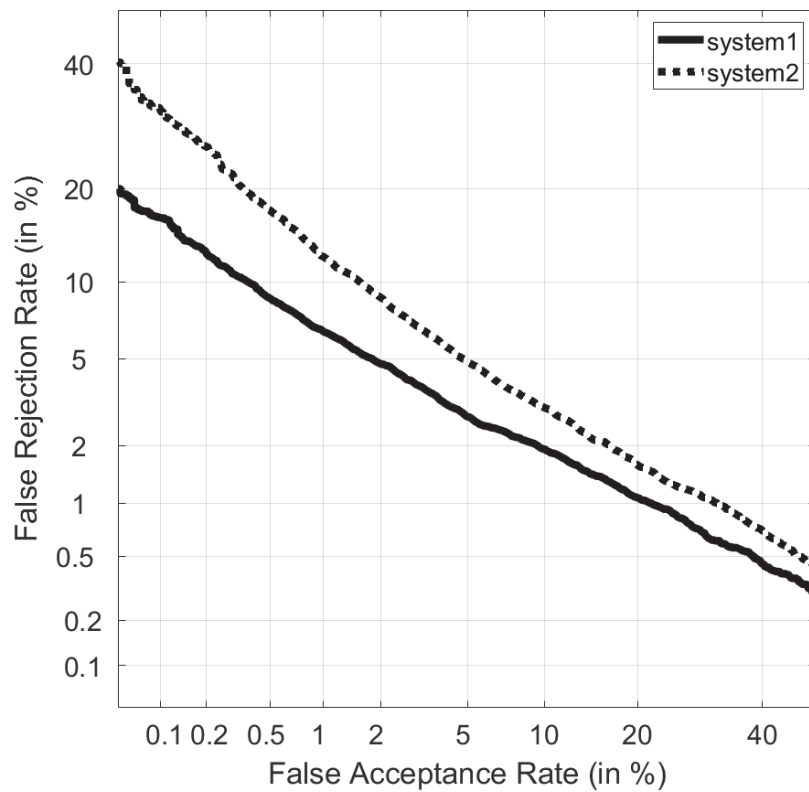
$\|a\| \times \|b\|$ is just the product of the magnitudes (norms) of a and b

Tip: the following norms have been calculated for you

$$\|db1\| = \sqrt{178} = 13.3417 \qquad \|db2\| = \sqrt{145} = 12.0416$$

Exercise 6. (4.5p)

The figure below shows the relation between the two types of errors (FAR and FRR) for two different biometric systems working in verification mode.



- a) Indicate the value of the EER of the two systems shown in the figure (system1 and system2) (mark it in the plot as well), From the EER, justify which of the two systems is better **(2p)**

- b) We want to operate the biometric systems as a convenience system. For this purpose, we select a decision threshold to make that only 2% of the genuine users are rejected. At this threshold, we would like that no more than 10% of the impostors are accepted. Indicate if any of the two systems fulfill the requirement. **(2.5p)**